141. METHANOGENIUM MEDIUM

KCl	0.34	g
$MgCl_2$ x 6 H_2O	4.00	g
$MgSO_4$ x 7 H_2O	3.45	g
NH_4Cl	0.25	g
$CaCl_2$ x 2 H_2O	0.14	g
K_2HPO_4	0.14	g
NaCl	18.00	g
Trace elements (see below)	10.00	ml
Vitamin solution (see below)	10.00	ml
$Fe(NH_4)_2(SO_4)_2$ x 7 H_2O	2.00	mg
$NaHCO_3$	5.00	g
Na-acetate	1.00	g
Yeast extract (Difco)	2.00	g
Trypticase (BBL)	2.00	g
Resazurin	1.00	mg
Cysteine-HCl x H_2O	0.50	g
Na_2S x 9 H_2O	0.50	g
Distilled water	1000.00	ml

Prepare the medium anoxically under 80% H_2 + 20% CO_2 gas atmosphere. For incubation use the same gas mixture at two atmospheres of pressure. If the medium is being used without gas mixture overpressure then adjust pH with a little hydrochloric acid. Final pH should be 7.5 for strain DSM 15219, DSM 16458 and DSM 18860; 7.0 for strains DSM 1497, DSM 1537, DSM 2067, DSM 2279, DSM 2373, DSM 17251 and DSM 17508; 6.8 for strain DSM 2095; and 6.5 for strain DSM 1498, DSM 15558 and DSM 22353.

Trace element solution:

Nitrilotriacetic acid	1.50	g
$MgSO_4$ x 7 H_2O	3.00	g
$MnSO_4$ x H_2O	0.50	g
NaCl	1.00	g
$FeSO_4$ x 7 H_2O	0.10	g
$CoSO_4$ x 7 H_2O	0.18	g
$CaCl_2$ x 2 H_2O	0.10	g
$ZnSO_4$ x 7 H_2O	0.18	g
$CuSO_4$ x 5 H_2O	0.01	g
$KAl(SO_4)_2$ x 12 H_2O	0.02	g
H_3BO_3	0.01	g

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Na_2MoO_4 x 2 H_2O	0.01	g
$NiCl_2$ x 6 H_2O	0.03	g
Na_2SeO_3 x 5 H_2O	0.30	mg
Distilled water	1000.00	ml

First dissolve nitrilotriacetic acid and adjust pH to 6.5 with KOH, then add minerals. Final pH 7.0 (with KOH).

Vitamin solution:

Biotin	2.00	mg
Folic acid	2.00	mg
Pyridoxine-HCl	10.00	mg
Thiamine-HCl x 2 H_2O	5.00	mg
Riboflavin	5.00	mg
Nicotinic acid	5.00	mg
D-Ca-pantothenate	5.00	mg
Vitamin B_{12}	0.10	mg
p-Aminobenzoic acid	5.00	mg
Lipoic acid	5.00	mg
Distilled water	1000.00	ml

For DSM 2373 increase the amount of trypticase to 6 g/l.

For DSM 2831 prepare the medium under 80% N_2 and 20% CO_2 gas atmosphere. After sterilization add 5.0 g/l sucrose from a sterile, anoxic stock solution prepared under N_2. The final pH should be 7.0.

For DSM 4254 add a filter-sterilized, anoxic solution of L-histidine to a final concentration of 80 mg/l.

For DSM 14042 prepare the medium under 80% N_2 and 20% CO_2 gas atmosphere. After sterilization add 0.05% methanol from a sterile, anoxic stock solution prepared under N_2. For incubation use the same gas mixture at two atmospheres of pressure. The final pH should be 7.0.

For DSM 15558 and DSM 16458 supplement medium after autoclaving with 0.5 g/l coenzyme M (mercaptoethanesulfonic acid) added from a filter-sterilized anoxic stock solution prepared under N_2. Use only one atmosphere overpressure of 80% H_2 and CO_2.

For DSM 21626 reduce the amount of NaCl to 6 g/l.

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